

C O T I V I T I

Clinical Natural Language Technology for Health Care

NLP · OCR · Computer Vision · LLM · LMM

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About Me



ECE Honors + Mathematics

Statistical reasoning, model reliability, systems thinking



Finance Minor

ROI framing, risk exposure, competitive positioning



Builder Mindset

I shipped a working POC for this assessment, not just a slide



Healthcare AI Focus

Applied AI where data quality has real human consequences

Why Cotiviti?

The opportunity to use data, analytics, and technology to improve healthcare accuracy, efficiency, and trust.



Operates where data quality directly affects both patient outcomes AND financial integrity. Most companies optimize for one.



Responsible AI philosophy: AI amplifies expert judgment, never replaces it. Confirmed in Cotiviti's CCC25 leadership content.



Post-Visit Review already deploys the NLT stack I researched. This isn't hypothetical for Cotiviti, it's core business.



#1 highest-designated leader, Everest Group Payment Integrity PEAK Matrix 2025. Market recognizes the differentiation.

What Is Clinical NLT?

A stack of technologies that lets machines read, interpret, and act on clinical language. Each layer solving a different part of the problem.



NLP

Parses physician notes, extracts diagnoses, handles negations ("no history of diabetes" vs "history of diabetes"), maps to ICD-10 & SNOMED CT.



OCR

Converts paper records, fax-based claims, and handwritten notes into machine-readable text. This is the prerequisite for everything above.



CV

Extracts clinical signal from radiology images, scanned operative reports, and multi-page PDF forms alongside text.



LLM/LMM

Reasons across text + images in a single inference call. Few-shot extraction, summarization, Q&A. No task-specific retraining required.

Healthcare raises the stakes: a single misread negation in a physician note carries direct financial and compliance consequences.

Relevant Trends

\$5.2B

to

\$16.0B

25.3% CAGR · 2025–2030

*NLP in Healthcare & Life Sciences
(MarketsandMarkets, 2025)*



Three Generational Shifts

Rule-based (cTAKES, MedLEE) → Transformer models (BioBERT, ClinicalBERT) → Instruction-tuned LLMs (Med-PaLM 2, BioMistral). Each shift dramatically expanded what AI can do with clinical text.



Ambient Clinical Intelligence

Voice-enabled LLMs generate EHR notes from patient-provider conversations in real time. Microsoft Dragon Copilot (Mar 2025) cut documentation burden by an estimated 40–70%. This is the fastest-growing ACI deployment to date.



NLP in Payment Integrity (Cotiviti's Space)

AI tools now cross-reference encounter data against HCC codes at claim submission. This shows the shift from retrospective audit to prospective accuracy. Cotiviti's Post-Visit Review operationalizes this today at enterprise scale.

Opportunities & Threats

OPPORTUNITIES



Unstructured Data Value

~80% of healthcare data is unstructured. LLMs unlock this for HCC coding, payment accuracy, quality measurement, and population health. These are core use cases.



Multimodal Reasoning

LLMs jointly process notes, labs, and imaging in one inference call, enabling holistic documentation audits that no prior technology could support.



Real-Time Coding Accuracy

NLP-assisted HCC and CPT validation at claim volume far exceeds manual audit team consistency, directly improving payer financial performance.



Interoperability via FHIR

FHIR-native NLP pipelines harmonize fragmented EHR systems. This is a capability Cotiviti significantly expanded through its 2025 Edifecs acquisition.

THREATS



Hallucination & Liability

LLMs produce confident but factually incorrect output. In payment integrity, a fabricated audit finding carries real legal and financial consequences.



HIPAA & PHI Compliance

Routing PHI through cloud-hosted LLMs introduces significant regulatory risk, often requiring costly on-premise or private-cloud deployments.



Model Bias

Models trained on academic center data underperform on rural or underserved populations, introducing systematic errors in risk scores and quality measures.



Regulatory Uncertainty

FDA and CMS frameworks for AI-assisted clinical coding tools are still evolving.

Strategic Recommendations for Cotiviti

Two investments extending Cotiviti's existing advantage:



01 | Fine-Tuned Clinical LLM for Prospective Documentation Gap Detection

Fine-tune a domain-specific LLM on Cotiviti's proprietary corpus of encounter data, HCC coding outcomes, and audit decisions. This model identifies documentation gaps at claim submission, shifting the value proposition from retrospective audit to prospective accuracy. The Edifecs FHIR pipeline provides the data infrastructure. No competitor without Cotiviti's data depth could replicate this product.



02 | Responsible AI Framework as a Market Differentiator

Cotiviti's leadership stated at CCC25 that AI should amplify expert judgment, not replace it. Formalizing this as a published Responsible Clinical AI framework - covering model explainability, clinician review checkpoints, bias monitoring across patient subgroups, and audit trails - directly addresses health plans' growing liability concerns. In a market crowded with AI vendors making broad promises, this makes Cotiviti the trusted choice and reinforces its values: transparency, authenticity, and accountability.

Proof of Concept: Clinical Report AI Assistant

A working Streamlit app that demonstrates the exact documentation-quality problem Cotiviti's Post-Visit Review solves at enterprise scale



Clinical Summary

Gemini 2.5 Flash Lite summarizes any pasted or uploaded clinical note into structured bullet points covering presenting issue, history, tests, treatment, and plan.



Entity Extraction

Extracts diagnoses, symptoms, labs, medications, imaging, providers, and time references from free text into a clean structured table.



Documentation Gaps

Flags missing dosages, ambiguous wording, absent test values, vague follow-up timing, and missing allergy documentation. This is exactly what coders and reviewers need.



Clinical Q&A

Answer any question about the note, grounded in the source document only. Next step: citations back to exact source lines for full auditability.

Tech Stack:

Python

Streamlit

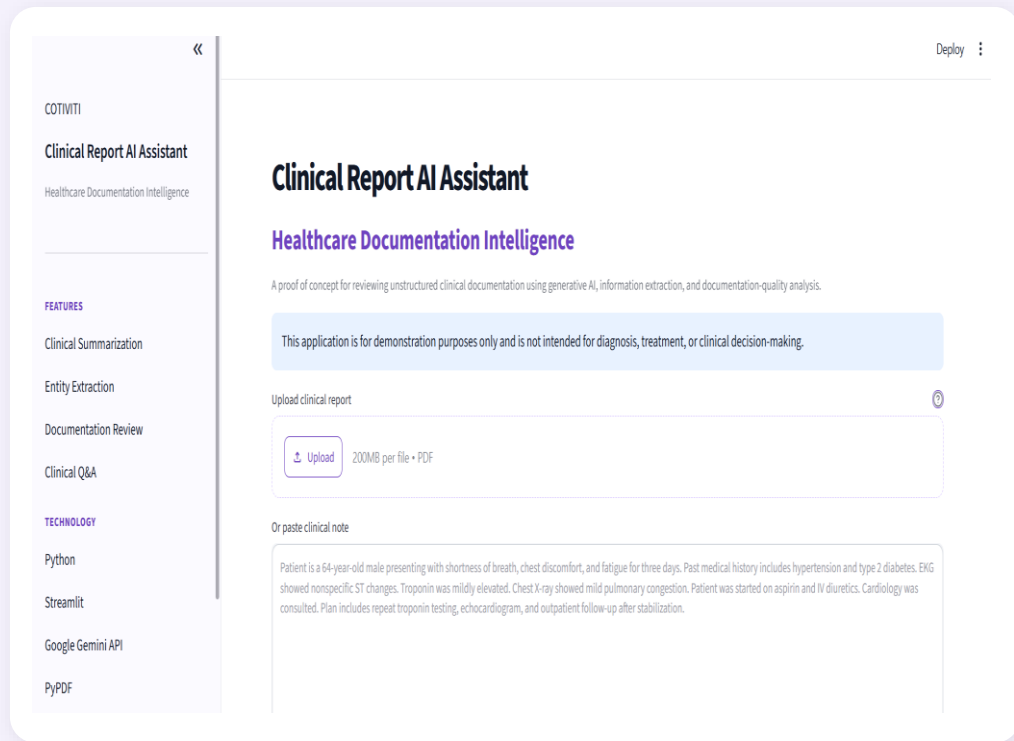
Gemini 2.5 Flash

PyPDF

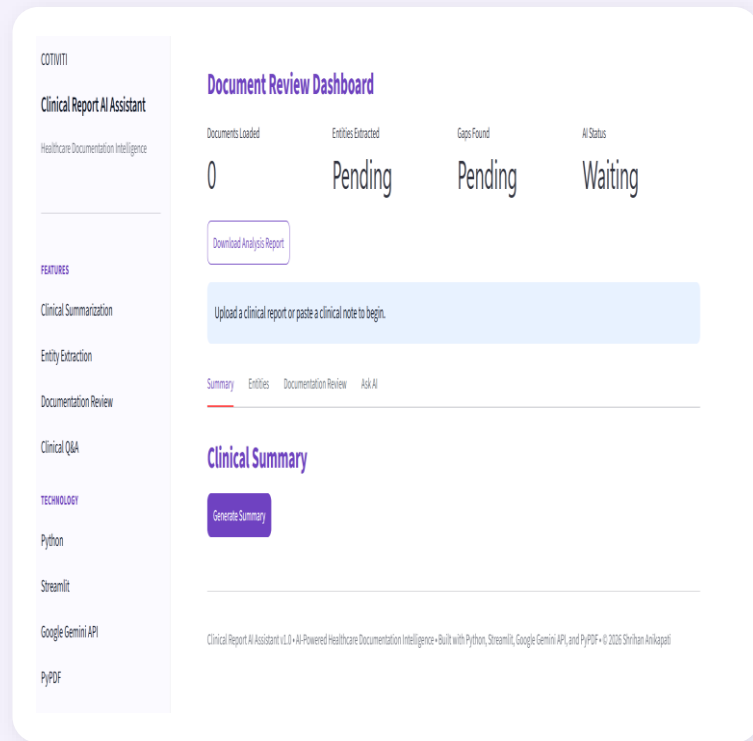
Pandas

Demo Mode

POC Demo — Live Interface



Upload PDF or paste clinical note and the text is extracted automatically



Real-time dashboard: Documents Loaded · Entities · Gaps · AI Status

Demo Summary

Each tab solves a different reviewer problem — the same problems Cotiviti's clients face at enterprise scale

01



Generate Summary

Paste or upload the clinical note. One click generates a structured bullet summary: presenting issue, PMH, tests, treatments, plan. Orients the reviewer in seconds vs. reading the full document.

Faster chart review for risk adjustment and quality teams.

02



Extract Entities

Structured table: conditions, symptoms, labs/tests, imaging, medications, provider, time references. Turns free text into machine-comparable facts.

Connects to HCC logic, CPT validation, and quality measures.

03



Documentation Review

Flags missing dosages, absent test values, vague follow-up, unclear allergy status, missing final assessment. Returns reviewer follow-up questions.

Human-in-the-loop: AI surfaces gaps, reviewer decides.

04



Ask AI

Grounded Q&A from the source document only. Answers stay within the clinical note, so hallucination is structurally limited. Next step: citations back to exact source lines.

Auditable AI: every answer traceable to a source document.

Thank You

Key Takeaways

- Clinical NLT is a \$16B market growing at 25% CAGR, and Cotiviti is already at the center of its most valuable application: payment integrity and risk adjustment.
- Two strategic investments: a fine-tuned clinical LLM and a published responsible AI framework, both extending what Cotiviti already does.
- My POC demonstrates the documentation-gap detection capability at the prototype level.
- ECE Honors + Mathematics + Finance Minor: I build the systems, understand the statistics, and frame the business value. Ready to contribute from day one.